

ROROS, Norway

WMO: 012880

Lat: 62.577N

Lon: 11.352E

Elev: 630

StdP: 93.98

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -31.7	(c) -28.2	(d) -34.8	(e) 0.2	(f) -31.7	(g) -30.9	(h) 0.2	(i) -28.1	(j) 12.9	(k) -3.4	(l) 11.5	(m) -3.5	(n) 0.4	(o) 320	(p) 0.754

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	23.6	14.3	21.3	13.5	19.2	12.7	15.6	20.9	14.6	19.3	13.5	18.0	3.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.5	10.4	17.0	12.5	9.7	16.1	11.5	9.1	15.2	45.5	20.7	42.7	19.4	39.9	17.9	19.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.2	8.7	7.5	-34.0	26.4	4.6	1.9	-37.3	27.8	-39.9	28.9	-42.5	30.0	-45.8	31.4
			-33.6	17.1	4.7	1.2	-37.0	18.0	-39.8	18.7	-42.4	19.4	-45.9	20.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-10.2	-8.4	-5.3	0.4	5.2	9.2	12.1	10.8	7.0	1.2	-4.1	-8.3
	DBStd	9.24	7.87	6.67	6.12	3.81	3.97	3.52	3.12	2.89	3.04	3.99	5.66	7.62
	HDD10.0	3518	626	515	473	287	159	57	14	25	98	274	424	568
	HDD18.3	6383	884	748	731	537	408	274	194	234	341	532	674	826
	CDD10.0	178	0	0	0	0	9	32	79	49	7	1	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	68	0	0	0	0	4	18	41	5	0	0	0	0
	CDH26.7	5	0	0	0	0	0	1	4	0	0	0	0	0
(14) Wind	WSAvg	3.0	3.0	3.0	3.3	3.1	3.4	3.4	3.0	2.7	3.0	2.8	2.6	2.8
(15) Precipitation	PrecAvg	621	47	40	36	28	46	67	84	85	54	47	44	42
	PrecMax	834	109	67	73	63	79	123	138	144	98	74	101	83
	PrecMin	490	6	9	11	12	14	27	31	29	21	16	8	18
	PrecStd	71	25	17	15	12	17	25	31	29	24	16	26	17
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	4.7	8.4	14.5	23.3	25.5	27.3	24.0	19.0	13.4	9.0	4.9
		MCWB	2.2	2.0	3.6	8.1	12.4	13.5	16.7	15.7	12.5	10.0	6.4	2.9
	2%	DB	2.5	2.8	5.8	11.7	19.4	22.8	24.2	21.1	16.2	10.6	6.3	3.7
		MCWB	1.0	0.9	2.6	6.1	11.2	13.3	15.0	13.7	11.6	7.6	4.5	2.0
	5%	DB	1.1	1.7	3.9	9.1	16.6	19.7	21.9	19.2	14.5	9.0	4.5	2.6
		MCWB	-0.3	0.0	1.4	4.7	10.1	12.1	14.2	13.3	10.6	6.5	2.9	1.0
	10%	DB	-0.3	0.3	2.6	7.0	13.3	17.1	19.4	17.2	12.9	7.4	2.7	1.1
		MCWB	-1.4	-1.1	0.5	3.4	8.1	11.0	13.2	12.5	9.8	5.5	1.4	-0.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	2.5	4.2	8.4	13.9	15.2	17.8	16.1	13.6	11.0	6.8	3.2
		MCDB	3.5	4.4	6.9	13.7	20.2	20.7	24.1	20.8	16.9	12.6	8.9	4.4
	2%	WB	1.1	1.2	3.0	6.5	12.1	14.0	16.2	15.1	12.4	8.1	4.7	2.1
		MCDB	2.5	2.6	5.8	10.7	18.8	20.6	22.5	19.4	15.1	10.1	6.4	3.5
	5%	WB	-0.1	0.0	1.7	5.0	10.4	12.9	15.0	14.0	11.2	6.8	3.0	1.1
		MCDB	1.0	1.6	3.7	8.7	15.6	18.9	20.1	17.8	13.8	8.6	4.2	2.3
	10%	WB	-1.4	-1.1	0.6	3.6	8.7	11.6	14.0	13.0	9.8	5.6	1.5	-0.1
		MCDB	-0.4	0.2	2.4	6.7	13.2	16.5	18.5	16.5	12.3	7.4	2.5	1.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.5	10.9	9.8	10.2	10.6	11.0	10.6	9.5	7.7	7.4	9.0
		MCDBR	7.2	8.2	9.9	13.4	17.2	17.1	16.4	14.7	12.9	10.4	8.1	9.2
	5% WB	MCWBR	6.3	6.9	7.3	8.5	9.5	8.4	8.2	8.3	8.3	7.6	6.7	8.0
		MCWBR	7.0	7.9	9.1	12.7	15.2	15.1	14.5	12.1	11.6	9.3	7.9	8.9
(40) Clear-Sky Solar Irradiance	taub		0.200	0.256	0.288	0.318	0.332	0.328	0.340	0.329	0.300	0.282	0.212	0.176
		taud	2.042	2.215	2.212	2.230	2.397	2.473	2.452	2.472	2.542	2.442	2.102	1.954
	Ebn at Noon		484	710	818	865	886	896	873	853	820	687	456	298
		Edh at Noon	32	61	90	110	102	97	96	87	66	49	29	18
(44) All-Sky Solar Radiation	RadAvg		0.23	0.88	2.37	4.23	4.90	4.90	4.61	3.56	2.24	0.99	0.30	0.12
	RadStd		0.03	0.10	0.20	0.27	0.28	0.42	0.48	0.34	0.21	0.13	0.04	0.01

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	-142	-178	N/A	N/A

Nomenclature: See separate page